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ANIRUDDHA BELSARE

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EDUCATION

2008 - 2013	Ph.D., Wildlife Science, University of Missouri, Columbia, Missouri, USA (with Dr. Matthew Gompper). *Dissertation title:* Disease ecology of free-ranging dogs in central India: Implications for wildlife conservation.
1991-1996	B.V.Sc. & A.H. (Bachelor of Veterinary Science & Animal Husbandry), Bombay Veterinary College, India.

Short courses

2021	Interactive Web-Based Visualizations and Decision Support Tools in Shiny/R for Quantitative Scientists. Workshop organized by the National Socio-Environmental Synthesis Center (SESYNC).
2018	Winter School on Agent-based Modeling of Social-Ecological Systems, Arizona State University, Tempe, Arizona.
2016	Individual-based (agent-based) modeling at Humboldt State University, Arcata, California.
2015	Designing courses for significant learning. Dee Fink and Associates.
2015	Preparing Future Faculty, Fall 2015 & Spring 2016. University of Missouri, Columbia.
2015	Entering Mentoring workshop series, Spring 2015. University of Missouri, Columbia.

RESEARCH EXPERIENCE

2023 – current	**Assistant Professor of Disease Ecology**, Auburn University College of Forestry, Wildlife & Environment and AU College of Veterinary Medicine. A major focus of my research group is applied epidemiology, utilizing virtual experimentation with agent-based modeling frameworks to inform real-world surveillance and control strategies for diseases such as chronic wasting disease (CWD), COVID-19, leptospirosis, Kyasanur Forest Disease. We are also investigating anthelmintic resistance at the livestock-wildlife interface. Another key focus of our research is free-roaming dog populations and related diseases.
2021 - 2023	**Associate Academic Research Scientist**, Department of Biology, Emory College of Arts and Sciences, Emory University, Atlanta, GA. My research focused on diseases of wild/free- ranging species including chronic wasting disease (white-tailed deer, reindeer), SARS-CoV-2 (white-tailed deer), leptospirosis, rabies, & chytridiomycosis.

2021 – 2022	**Faculty Affiliate**, Wildlife Biology Program, University of Montana. I continue the chronic wasting disease modeling work in collaboration with University of Montana faculty and State Wildlife Agencies.
2019 - 2021	**Research Associate**, Boone & Crockett Quantitative Wildlife Center, Michigan State University, East Lansing, MI. I developed new, high-level quantitative tools that provide decision support while addressing the challenges of wildlife conservation. I assessed the impact of alternate harvest strategies on chronic wasting disease spread and persistence in regional white-tailed deer populations.
2017 - 2019	**Postdoctoral Fellow**, Center for Modeling Complex Interactions, University of Idaho, Moscow ID, USA. Leader, OneHealth Working Group: Collaborative research on host-pathogen systems of public health or conservation concern. a) Canine rabies project (with Dr. Craig Miller, CMCI), b) Leptospirosis modeling project (with Dr. Claudia Munoz-Zanzi, University of Minnesota), and c) Raccoon roundworm management project (with Dr. Matt Gompper, University of Missouri). Collaborator & Modeler, Modeling Access Grant Project: Bighorn sheep pneumonia modeling project (with Dr. Ryan Long & Dr. Frances Cassirer).
2014 - 2017	**Postdoctoral Fellow**, School of Natural Resources, University of Missouri, Columbia, USA. My research focused on the analysis of wildlife diseases in the state of Missouri. I developed models to understand the spread of chronic wasting disease in white-tailed deer in Missouri and provide a framework for designing effective surveillance strategy.
2008 - 2013	**Ph.D.**, Department of Fisheries and Wildlife Sciences, University of Missouri, Columbia, USA. My dissertation research combined ecologic, epidemiologic and model-based investigations to study the disease ecology of free-ranging dogs around a protected area in central India. I examined the local and global implications of dog diseases and disease control measures for wildlife conservation.
2007 - 2009	Indo-Norwegian project on wildlife-human conflict. Rabies as a driver of Human-Wolf conflict and the role of free ranging domestic dogs as carriers of the disease. Maharashtra, India.
2006 - 2007	University of Missouri and Rufford Small Grants. Fox ecology project and survey of disease prevalence in free-ranging domestic dogs and possible spillover risk for wildlife.
2004 - 2006	Maharashtra Forest Department and Wildlife Trust of India, New Delhi. Helping the Maharashtra State Forest Department to rescue or treat endangered wild carnivores. http://www.projectwaghoba.in/docs/athreya_rap2004_final_report.pdf
2003	Ecollage, Pune and Wildlife Protection Society of India, New Delhi. Study of the man-leopard conflict in the Junnar Forest Division, Pune District, Maharashtra.

VETERINARY PROFESSIONAL EXPERIENCE

2015	Expert Veterinarian, King Cobra Telemetry Project, Central Kalimantan, Indonesia. Surgically implanted radio-transmitters in two Sumatran Spitting cobras (*Naja sumatrana*) and one Reticulated Python (*Python reticulatus*) (supported by Copenhagen Zoo, Denmark).
2009 - 2010	Curricular Practical Training: After Hours Clinical Crew, Veterinary Medical Teaching Hospital, University of Missouri, Columbia, Missouri, USA.
2001 - 2012	Veterinary Consultant, Maharashtra Forest Department. Provided veterinary support during wildlife emergencies, rescue and treatment of wild animals and birds.
1999 - 2008	Veterinary Practitioner. Owner of private clinic in Pune, India. Medical and surgical treatment of pets (dogs, cats and birds) and livestock in and around Pune city.
2008	Consultant Veterinary Surgeon, King Cobra Telemetry Project, National Geographic Society and Agumbe Rainforest Research Station (with Romulus Whitaker and Matt Goode).
2005 - 2008	Veterinary Consultant, Madras Crocodile Bank Trust, Chennai, India.
2001 - 2004	Veterinary Officer, Rajiv Gandhi Zoological Park & Wildlife Research Center, Pune. Healthcare and management of zoo animals were two main responsibilities. I implemented controlled breeding programs in prolifically breeding species using MGA implants and also started using transponder microchips for better management of zoo animals.
1996 - 2001	Consultant Veterinarian, Wild Animal Orphanage & Pune Snake Park.

TEACHING AND MENTORING EXPERIENCE

Teaching

Auburn University

2023 - current	Disease Ecology (WILD 5200/6200), One Health Connections (WILD 5440/6440), Wildlife Rabies: A One Health Approach (WILD 7420), Tools and Challenges for the Modern Scientist (CRN 18603 / WILD 7450), Wildlife Diseases (VMED 9840-001), Introduction to Public Health (FOWS 7400). Fall 2023: Taught One Health Connections at the Ventress Correctional Facility for the Alabama Prisons Arts + Education Project.

UKCEH

2021-2022	Invited speaker for the UK Center for Ecology & Hydrology training course ‘One Health, data and models for zoonotic disease management’.

Purdue University

2021 (Spring)	Instructor, Readings in Disease Ecology (FNR 59800).

Michigan State University

2021	Guest lecturer, Lyman Briggs College. Introductory Biology.

2019, 2020	Guest lecturer, Wildlife Disease Ecology (FW463).
	Guest lecturer, Wildlife Disease Ecology and Management Veterinary Clerkship (MSU LCS 610).
University of Idaho	
2017-2019	Guest lecturer, Ecology of Terrestrial Vertebrates (WLF314).
	Guest lecturer, Wildlife Management (WLF492).
University of Missouri	
2016, 2014 (Fall)	Instructor, Wildlife Disease Ecology (FW 4810/7810): I co-taught this undergraduate/graduate-level course with Dr. Matthew Gompper.
2016 (Spring)	Instructor, Animal Population Dynamics and Management (FW 4500/7500).
2012 - 2014	Instructor, General Biology Laboratory (BIO 1020): Fall 2012, Spring 2013 and Fall 2013, Fall 2014.
2010 (Spring)	Teaching Assistant, Introductory Zoology with Laboratory (FW 1100).
State Forest Departments, India	
2005 - 2006	Designed and taught short courses in chemical immobilization of wild animals for forest department personnel and veterinarians from seven states in India. Rufford Small Grants funded this project.
Maharashtra Forest Department, India	
2003 - 2004	Trained five emergency response teams (Maharashtra Forest Department personnel) to better manage human-leopard conflict.

Mentoring

Auburn University	Major advisor for Kayla Alston, Lauren Wakefield, Hunter Holcomb and Amulya Malyala (all MS candidates).
University of Illinois	
Chicago	Serving on the dissertation committee of Soraida Garcia (PhD candidate, Biological Sciences)
Purdue University	
2021 - 2024	Served on the dissertation committee of Jonathan Brooks (Ph.D. candidate, Department of Forestry and Natural Resources).
University of Minnesota	
2015 - 2016	Advised Meghan Mason, a Ph.D. candidate in the School of Public Health, on her agent-based modeling work.
University of Missouri	
2015	Lucy Mills, graduate student in Department of Agricultural Economics. Agent-based model of collective entrepreneurship.
2014	Anna Maness, undergraduate student in Fisheries and Wildlife Science. Raccoon roundworm project.

GRANTS, FELLOWSHIPS AND AWARDS

2024 – 2025	Co-Principal Investigator, Monitoring Alabama white-tailed deer populations. Alabama Department of Conservation and Natural Resources ($107,124).
2024 - 2025	Principal Investigator, Optimizing the surveillance and management of CWD on tribal lands for the preservation of culture and food sovereignty. USDA – APHIS (Subaward through University of Minnesota $55,272).
2023 – 2024	Principal Investigator, Improving chronic wasting disease surveillance of captive deer facilities in Texas using a model-informed adaptive management approach. USDA – APHIS ($103,976).
2022 – 2025	Principal Investigator, Model-informed chronic wasting disease management: Assessing relative risk of chronic wasting disease transmission from public activities. Safari Club International Foundation ($279,790).
2021 – 2022	Principal Investigator, Modeling Chronic Wasting Disease in Missouri. Missouri Department of Conservation (Subaward through University of Montana $74,998).
2020 - 2023	Project Partner, IndiaZooRisk+: Using OneHealth approaches to understand and co-develop interventions for zoonotic diseases affecting forest communities in India. UK Research and Innovation Global Challenges Research Fund Health and Context Call 2019.
2019 - 2021	Co-Principal Investigator, Optimizing CWD Surveillance: Regional synthesis of demographic, spatial and transmission-risk factors. Michigan Department of Natural Resources and Michigan State University Joint Wildlife Disease Initiative ($241,094, PI Krysten Schuler).
2018	Project Director, Canine rabies genome sequencing project. Sudden Opportunity Grant (CMCI, University of Idaho) ($18000).
2016 - 2021	Co-Principal Investigator, 'Bringing "OneHealth" to rabies research in India: integrating animal ecology, disease ecology, and human health'. Wellcome Trust/ DBT India Alliance Fellowship ($35000, PI Abi Vanak).
2013, 2010	Annual Research and Creative Activities Forum, Graduate Professional Council, University of Missouri. 2nd place (2013) and 1st place (2010) in Veterinary Medicine/Medicine/Health Sciences.
2012 - 2013	Conservation Biology Fellowship, University of Missouri ($3000).
2008 - 2013	G. Ellsworth Huggins Scholarship, University of Missouri ($29500 per year).
2006	Rufford Small Grant for Nature Conservation (£5000); co-investigator for project titled 'Survey of disease prevalence in free-ranging domestic dogs around the Great Indian Bustard Sanctuary, India (with A. T. Vanak).
2005	Rufford Small Grant for Nature Conservation (£5000) for project titled 'Standardization of procedures required for dealing with wildlife emergencies by training veterinarians and Forest Department personnel in states with high human-wildlife conflicts in India'.

PUBLICATIONS

1. Wakefield, L., Cain, A., Cerny, C., Reed, H., & Belsare, A. V. (2026). A simulation-based approach to strengthen chronic wasting disease surveillance in captive cervid populations. *PLoS One*, In Press.

2. Brooks, J. D., A. V. Belsare, J. N. Caudell, and P. A. Zollner. 2026. Harvest increase and culling as tools for managing chronic wasting disease in white-tailed deer. *Journal of Wildlife Management,* e70211. https://doi.org/10.1002/jwmg.70211

3. Rupprecht, C. E., & Belsare, A. V. (2026). Rabies and Pinnipeds Reviewed: Premonitions, Perturbations, and Projections? *Veterinary Sciences*, 13(2), 200. https://doi.org/10.3390/vetsci13020200

4. Ditchkoff, S. S., and A. V. Belsare. 2025. Much ado about modeling: Why fieldwork should remain the soul of wildlife ecology and management. *Wildlife Society Bulletin* e1608.https://doi.org/10.1002/wsb.1608

5. Willoughby, J.R., Lamka, G.F., Dunning, K.H., Narine, L., Belsare, A.V. and Sundaram, M. Using AI enhanced agent-based models to support management of wild populations. *Landscape Ecology* **40**, 129 (2025). https://doi.org/10.1007/s10980-025-02149-2

6. Rupprecht C. E., Belsare A. V., Cliquet F, Mshelbwala P. P., Seetahal JFR, Wicker V. V. The Challenge of Lyssavirus Infections in Domestic and Other Animals: A Mix of Virological Confusion, Consternation, Chagrin, and Curiosity. *Pathogens*. 2025 Jun 13;14(6):586. doi: 10.3390/pathogens14060586

7. Belsare, A. V., Briggs, D.J., Gongal, G., Mani, R.S. and Rupprecht, C.E. (2025), Enhancing Human Post-Exposure Rabies Prophylaxis in Canine Rabies Endemic Regions With an Interactive Web Solution. *Public Health Challenges*, 4: e70030. https://doi.org/10.1002/puh2.70030

8. Rupprecht, C. E. and Belsare, A. V. "Emerging Zoonotic and Wildlife Pathogens: Disease Ecology, Epidemiology and Conservation," *Journal of Wildlife Diseases* 61(1), 281-282, (31 January 2025). https://doi.org/10.7589/0090-3558-61.1.BR1

9. Belsare, A. V., Gompper, M. E., Mason, M., and Munoz-Zanzi, C. (2022). Investigating *Leptospira* dynamics in a multi-host community using an agent-based modeling approach. *Transboundary and Emerging Diseases*, 00, 1-10. https://doi.org/10.1111/tbed.14750

10. Owen, J. C., Landwerlen, H., Dupuis, A. P., Belsare, A. V., Sharma, D., Wang, S., Ciota, A.T., and Kramer, L. (2021). Reservoir hosts experiencing food stress alter transmission dynamics for a zoonotic pathogen. *Proceedings of the Royal Society B*, 288: 20210881. https://doi.org/10.1098/rspb.2021.0881

11. Mysterud, A., Viljugrein, H., Rolandsen, C. M, and Belsare, A. V. (2021). Harvest strategies for the elimination of low prevalence wildlife diseases. *Royal Society Open Science*, 8: 210124. https://doi.org/10.1098/rsos.210124

12. Belsare, A. V., Millspaugh, J.J., Mason, J.R., Sumners, J., Viljugrein, H., and Mysterud, A. (2021). Getting in front of chronic wasting disease: Model-informed proactive approach for managing an emerging wildlife disease. *Frontiers in Veterinary Science*, 7, 1154. doi: 10.3389/fvets.2020.608235

13. Belsare, A. V., & Vanak, A. T. (2020). Modelling the challenges of managing free-ranging dog populations. *Scientific Reports*, 10(1). doi:10.1038/s41598-020-75828-6

PUBLICATIONS (continued)

14. Belsare, A. V., & Stewart, C. M. (2020). OvCWD: An agent-based modeling framework for informing chronic wasting disease management in white-tailed deer populations. Ecological Solutions and Evidence, 1(1), e12017. doi:10.1002/2688-8319.12017

15. Belsare, A. V., Gompper, M. E., Keller, B., Sumners, J., Hansen, L. P., & Millspaugh, J. J. (2020). Size matters: Sample size assessments for chronic wasting disease surveillance using an agent-based modeling framework. *MethodsX*, 7, 100953. doi: https://doi.org/10.1016/j.mex.2020.100953

16. Belsare, A. V., Gompper, M. E., Keller, B., Sumners, J. A., Hansen, L. P., & Millspaugh, J. J. (2020). An agent-based framework for improving wildlife disease surveillance: A case study of chronic wasting disease in Missouri white-tailed deer. *Ecological Modelling*, 417, 108919. doi: 10.1016/j.ecolmodel.2019.108919 (***F1000 Prime Recommended Article***).

17. Al-Warid, H. S., Belsare, A. V., Straka, K., Gompper, M. E., & Eggert, L. S. (2018). Genetic polymorphism of *Baylisascaris procyonis* in host infrapopulations and component populations in the Central USA. *Parasitology International*, 67(4), 392–396. doi: 10.1016/j.parint.2018.03.005

18. Richardson, D. J., Leveille, A., Belsare, A. V., Al-Warid, H. S., & Gompper, M. E. (2017). Geographic Distribution Records of *Macracanthorhynchus ingens* (Archiacanthocephala: Oligacanthorhynchidae) from the Raccoon, Procyon lotor in North America. *Journal of the Arkansas Academy of Science*, 71(1), 203–205.

19. Al-Warid, H. S., Belsare, A. V., Straka, K., & Gompper, M. E. (2017). *Baylisascaris procyonis* roundworm infection patterns in raccoons (*Procyon lotor*) from Missouri and Arkansas, USA. *Helminthologia*, 54(2), 113–118. doi:10.1515/helm-2017-0011

20. Belsare, A. V., & Gompper, M. E. (2015). To vaccinate or not to vaccinate: Lessons learned from an experimental mass vaccination of free-ranging dog populations. *Animal Conservation*, 18(3), 219–227. doi:10.1111/acv.12162 (***Highlighted article***)

21. Belsare, A. V., & Gompper, M. E. (2015). A model-based approach for investigation and mitigation of disease spillover risks to wildlife: Dogs, foxes and canine distemper in central India. *Ecological Modelling*, 296, 102–112. doi: 10.1016/j.ecolmodel.2014.10.031

22. Vanak, A. T., Belsare, A. V., Uniyal, M., & Ali, R. (2014). Science in the doghouse. *Current Science*, 107(3), 341–342.

23. Belsare, A. V., Vanak, A. T., & Gompper, M. E. (2014). Epidemiology of viral pathogens of free-ranging dogs and Indian foxes in a human-dominated landscape in central India. *Transboundary and Emerging Diseases*, 61(SUPPL1.), 78–86. doi:10.1111/tbed.12265

24. Belsare, A. V., & Vanak, A. T. (2013). Use of Xylazine Hydrochloride–Ketamine Hydrochloride for immobilization of Indian fox (*Vulpes bengalensis*) in field situations. *Journal of Zoo and Wildlife Medicine*, 44(3), 753–755. doi:10.1638/2012-0158r.1

25. Belsare, A. V., & Gompper, M. E. (2013). Assessing demographic and epidemiologic parameters of rural dog populations in India during mass vaccination campaigns. *Preventive Veterinary Medicine*, 111(1–2), 139–146. doi: 10.1016/j.prevetmed.2013.04.003

26. Belsare, A. V., & Athreya, V. R. (2010). Use of Xylazine hydrochloride-Ketamine hydrochloride for immobilization of wild leopards (*Panthera pardus fusca*) in emergency situations. *Journal of Zoo and Wildlife Medicine*, 41(2), 331–333. doi:10.1638/2009-0072R1.1

27. Athreya, V. R., & Belsare, A. V. (2008). Morphometry of leopards from Maharashtra, India. *Cat News*, 48 (Spring).

28. Athreya, V. R., Thakur, S. S., Chaudhuri, S., & Belsare, A. V. (2007). Leopards in human-dominated areas: a spillover from sustained translocations into nearby forests? *Journal of the Bombay Natural History Society*, 104(1), 45–50.

29. Belsare, A. V., Athreya, V. R., Thakur, S. S., & Chaudhuri, S. (2004). Life-long Identification Microchips in Leopards Caught in Conflict Areas in Maharashtra, India. *Cat News*, 41 (Autumn), 10–11.

30. Umrigar, K. D., & Belsare, A. V. (2003). Contraception in a Blackbuck (*Antilope cervicapra*) using melengesterol acetate. *Zoos' Print Journal*, 18(6), 1129. doi: 10.11609/jott.zpj.18.6.1129

Code and data:

1. Belsare, Aniruddha (2025, November 11). “CapOvCWD” (Version 1.1.0). CoMSES Computational Model Library. Retrieved from: https://doi.org/10.25937/87rx-af75
2. Belsare, Aniruddha; Owen, Jennifer (2021, March 11). “AMRO_CULEX_WNV” (version 1.1.0). CoMSES Computational Model Library. Retrieved from: https://doi.org/10.25937/hfcr-bt18
3. Belsare, Aniruddha (2021, February 4). anyadoc/ABMDataNorwayReindeerCWD: ABMDataNorwayReindeerCWD (Version V1.0). Zenodo. http://doi.org/10.5281/zenodo.4501249
4. Belsare, Aniruddha; Vanak, Abi (2020, August 01). “DOGPOPDY: ABM for ABC planning” (Version 1.0.0). CoMSES Computational Model Library. Retrieved from: https://doi.org/10.25937/9nge-4s45
5. Belsare, Aniruddha (2020, April 13). “MI*Ov*POPsurveillance” (version 1.0.0). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/fdke-rp28
6. Belsare, Aniruddha (2019, December 13). “MI*Ov*CWD” (Version 1.0.0). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/6qeq-1c13
7. Belsare, Aniruddha, Long, Ryan, Cassirer, E Frances (2019, November 05). “BHSPopDy (Bighorn sheep population dynamics)” (Version 1.2.0). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/a7fz-tw30
8. Belsare, Aniruddha (2019, September 18). “MI*Ov*POP” (version 1.1.0). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/kv07-3e08
9. Belsare, Aniruddha (2019, August 9). anyadoc/FranklinCWDsurveillance_Rcode: ABM framework output for Franklin County & R code for analysis (Version v.1.0.0). Zenodo. http://doi.org/10.5281/zenodo.3364607
10. Belsare, Aniruddha, Gompper, Matthew, Millspaugh, Joshua J (2019, August 08). “MO*Ov*POP” (Version 2.1.2). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/cnex-s628
11. Belsare, Aniruddha, Gompper, Matthew, Millspaugh, Joshua J (2019, August 08). “MO*Ov*POPsurveillance” (Version 2.1.2). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/8hpz-9y96
12. Belsare, Aniruddha, Mason, Meghan, Gompper, Matthew, Munoz-Zanzi, Claudia (2019, March 12). “MHMSLeptoDy (Multi-host, multi-serovar Leptospira Dynamics Model)” (Version 1.1.2). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/zp7k-bq40

PUBLICATIONS (continued)

13. Belsare, Aniruddha, Gompper, Matthew (2017, April 04). "DogFoxCDVspillover" (Version 1.1.0). *CoMSES Computational Model Library*. Retrieved from: https://doi.org/10.25937/167q-wr96

Manuscripts in preparation (available upon request)

1. Gompper, M. E., and Belsare, A.V. Intraguild predation benefits the killing species in communities with shared pathogens.
2. Belsare, A.V. Hematologic and serum biochemistry reference intervals for wild-caught leopards (*Panthera pardus fusca*) from Maharashtra, India.

Reports, popular articles, and unpublished materials

1. Belsare, A.V. 2020. Nebulizer therapy with isotonic saline to decrease aerosol transmission of SARS-CoV-2. eLetter in response to Prather, K. A. et al. "Reducing transmission of SARS-CoV-2" published in *Science*. https://science.sciencemag.org/content/early/2020/06/08/science.abc6197/tab-e-letters
2. Singh, M., I. Malik, W. Dittus, A. Sinha, A. Belsare, S. Walker, S. Molur, B. Wright, J. Lenin & S. Chaudhuri (2019). Action plan for the control of commensal, non-human primates in public places. Submitted to the Ministry of Environment & Forests, Government of India in 2005. *Zoo's Print* 34(11): 1–13.
3. Vanak, A.T. & Belsare, A.V. The street is no place for dogs. *The Hindu*, 3 October 2016.
4. Belsare, A.V. 2013. Diseases of free-ranging dogs: Implications for wildlife conservation in India. *Current Conservation* 7 (4): 3-12.
5. Belsare, A.V. 2012. To Dart or Not To Dart- Demystifying Wild Animal Immobilization. Conservation India. (http://www.conservationindia.org/resources/opinion/to-dart-or-not-to-dart-demystifying-wild-animal-immobilization)
6. Belsare, A.V. 2011. Rabies: a neglected killer. *Current Conservation* 5 (2): 19-21.
7. Vanak, A.T., Belsare, A.V. and Gompper, M.E. 2007. Survey of disease prevalence in free-ranging domestic dogs and possible spillover risk for wildlife. *Report submitted to the Rufford Small Grants Foundation*, *UK*. Pp 1-13.
8. Athreya, V.R. and Belsare, A.V. 2007. Human-leopard conflict management guidelines. Kaati Trust, Pune, India. (https://www.academia.edu/download/66462154/Human-Leopard_Conflict_Management_Guidel20210421-20673-1bcuh19.pdf)
9. Athreya, V.R. and Belsare, A.V. 2005. Helping the Maharashtra Forest Department rescue or treat endangered wild carnivores. *Report submitted to the Wildlife Trust of India, New Delhi and the Office of the Chief Wildlife Warden, Maharashtra*.
10. Athreya, V.R., Thakur, S.S., Chaudhuri, S., and Belsare, A.V. 2004. A study of man-leopard conflict in the Junnar forest division, Pune district, Maharashtra. *Report submitted to the Office of the Chief Wildlife Warden, Maharashtra State Forest Department, and the Wildlife Protection Society of India, New Delhi, India*.

ABSTRACTS AND MEETING PRESENTATIONS

2025	Analyzing Patterns of Gastrointestinal Nematode Infections at the Wildlife-Domestic Animal Interface. [Conference Presentation]. The 32nd Annual Conference of The Wildlife Society, Oct 6-9, 2025. Edmonton, Alberta, Canada. (Jones, K. & Belsare, AV)
2025	Optimizing Chronic Wasting Disease Surveillance in Captive Cervid Facilities: An Agent-based Modeling Approach. [Conference Presentation]. The 32nd Annual Conference of The Wildlife Society, Oct 6-9, 2025. Edmonton, Alberta, Canada. (Wakefield, L. & Belsare, AV)
2025	Bias in Wildlife Disease Investigations: Communications, Collaborations & Changes In Absentia. [Conference Presentation]. The 32nd Annual Conference of The Wildlife Society, Oct 6-9, 2025. Edmonton, Alberta, Canada. (Rupprecht, C. & Belsare, AV)
2025	Ecology, epidemiology and model-based explorations: An analytical approach for investigating wildlife disease systems. Distinguished Speaker Seminar, Archbold Biological Station, Venus, Florida. February 27, 2025.
2025	Getting in Front of CWD: Challenges and strategies for managing an emerging wildlife disease. Natural Resources Webinar Series, Alabama Cooperative Extension System. February 26, 2025.
2025	May the force (of infection) be with you: Model-based assessments of chronic wasting disease management strategies. [Conference Presentation]. 2025 Southeast Deer Study Group Meeting, 16-18 February 2025. Cambridge, Maryland.
2025	Optimizing Chronic Wasting Disease Surveillance: An agent-based modeling approach for captive deer facilities. [Conference Presentation]. 2025 Southeast Deer Study Group Meeting, 16-18 February 2025. Cambridge, Maryland. (Wakefield, L. & Belsare, AV)
2025	Novel Approaches to CWD Surveillance and Management: The Intersection of RT-QuIC Diagnostics and Epidemiological Modeling. [Conference Presentation]. 2025 Southeast Deer Study Group Meeting, 16-18 February 2025. Cambridge, Maryland. (Larsen, PA., Belsare, AV, Rowden, GR., Larsen, RJ., Anu Li, E., Grunklee, MK., Huang, MHJ., Karwan, DL., Lichtenberg, SL, Wolf, TM. and Schwabenlander, MD)
2024	Sustainable and Equitable Strategies for Human Rabies Elimination: The role of free-roaming dog population management in achieving the Zero by 30 goal. [**Keynote Speaker**]. XXXV International Conference on Rabies In The Americas 2024, Buenos Aires, Argentina. https://www.ritaconference.org/wp-content/uploads/2024/11/PDF-RITA-2024.pdf
2024	Agents on the Move: Simulating Spatiotemporal Dynamics of Hosts, Vectors and Pathogens in Agent-based Models. [Conference Presentation]. The 31st Annual Conference of The Wildlife Society, Oct 19-23,2024. Baltimore, Maryland. https://shorturl.at/DcUGt

ABSTRACTS AND MEETING PRESENTATIONS (continued)

2024	Effect of Web App Use on Hunter and Non-hunter Intention to Engage in Chronic Wasting Disease Mitigating Behaviors. [Conference Presentation]. The 31st Annual Conference of The Wildlife Society, Oct 19-23, 2024. Baltimore, Maryland. (Brooks, J., Belsare, A. V., Caudell, J., Ma, Z., & P. A. Zollner). https://www.xcdsystem.com/tws/program/0i3Xcy6/index.cfm?pgid=340&sid=10279&abid=29922
2024	The Use of Agent Based Modeling as a Means to Improve Chronic Wasting Disease Surveillance. [Poster presentation]. The 31st Annual Conference of The Wildlife Society, Oct 19-23, 2024. Baltimore, Maryland. (Wakefield, L. & Belsare, A. V.). https://www.xcdsystem.com/tws/program/0i3Xcy6/index.cfm?pgid=340&sid=10322&abid=30328
2024	A pragmatic agent-based modeling tool for improving chronic wasting disease surveillance of permitted deer breeding facilities in Texas. Technical presentation, 2024 Annual Conference Alabama Chapter of The Wildlife Society. July 23-25, 2024. Milbrook, Alabama. (Wakefield, L. & Belsare, A. V.).
2024	Investigating helminth communities of wild and domestic ruminants. Poster presentation, 2024 Annual Conference Alabama Chapter of The Wildlife Society. July 23-25, 2024. Milbrook, Alabama. (Alston, K & Belsare, A. V.).
2024	Investigating complex host-pathogen systems using pragmatic agent-based models. Invited speaker, RIDE Seminar Series, University of Minnesota College of Veterinary Medicine. April 10, 2024. St. Paul, Minnesota.
2023	Evaluating a novel chronic wasting disease control strategy using an agent-based modeling approach. Oral presentation, The Wildlife Society 30th Annual Conference. November 5 -9, 2023. Louisville, Kentucky. (Brooks, J., Belsare, A. V., Caudell, J., and Zollner, P.)
2023	Agents of change: Individual-based models to investigate complex zoonotic disease systems. Invited speaker, 11th Annual Phi Zeta Research Day, College of Veterinary Medicine, Tuskegee University. September 19, 2023. Tuskegee, Alabama.
2023	Investigating complex wildlife disease systems using an agent-based modeling approach. Invited speaker, Fall 2023 Warnell Speaker Series, Warnell School of Forestry and Natural Resources, University of Georgia. September 7, 2023. Athens, Georgia.
2023	Elucidating SARS-CoV-2 dynamics in white-tailed deer using an agent-based modeling approach. 2023 International Association of Landscape Ecology (IALE)- North America Annual Meeting. March 19 – 23,2023. Riverside, California. (with Dave Civitello).
2022	Simulating the spread of an emerging multihost pathogen in a novel wildlife reservoir. Contagion on Complex Social Systems Workshop 2022. University of Colorado Boulder. August 10-12, 2022. Boulder, Colorado.

ABSTRACTS AND MEETING PRESENTATIONS (continued)

2022	Effectiveness of artificial ecological trap for mitigating chronic wasting disease in white-tailed deer (*Odocoileus virginianus*). Annual Meeting of the American Society of Mammalogists, Tucson, Arizona. June 17-21, 2022. (Brooks, J., Belsare, A. V., Caudell, J., and Zollner, P.)
2020	ZeroBy30? Challenges to eliminating Rabies in India. Poster presented at the World One Health Congress. Oct 30 – Nov3, 2020. (with Abi Vanak, A. Kulkarni, A. Kumar, S. Sapre, N. Panchamiya, I. Banerjee).
2020	Decision support tool for CWD management. The Natural Resources Commission, Michigan. June 3, 2020.
2020	Modeling effects of deer harvest regulations on CWD. Fish and Wildlife Health Committee, Association of Fish and Wildlife Agencies. North American Wildlife and Natural Resources Conference. March 12, 2020. Omaha, Nebraska.
2020	Model-informed CWD Management. Science and Research Committee, Association of Fish and Wildlife Agencies. North American Wildlife and Natural Resources Conference. March 11, 2020. Omaha, Nebraska.
2019	Model-informed strategies for chronic wasting disease management. The Wildlife Society & American Fisheries Society 2019 Joint Annual Conference. September 29 – October 3, 2019. Reno, Nevada.
2017	Size matters: sample size calculations for harvest-based wildlife disease surveillance using an agent-based framework. 24th Annual Conference of The Wildlife Society. September 2017. Albuquerque, New Mexico.
2017	Raccoon roundworm intensity distributions across hosts and modeled implications for population management. 24th Annual Conference of The Wildlife Society. September 2017. Albuquerque, New Mexico. (Gompper ME & Belsare AV).
2017	Host biodiversity and tick-borne disease: Implications for temperate agroforestry (J.R. Falco, A.V. Belsare, M.E. Gompper, S. Jose). 15th North American Agroforestry Conference. June 2017. Blacksburg, Virginia.
2017	A model-based framework for improving chronic wasting disease surveillance in white-tailed deer populations of Missouri. 40th Annual meeting of the Southeast Deer Study Group. February 2017, St. Louis, Missouri.
2014	ABM for CWD: An adaptive disease management strategy. 38thAnnual Midwest Deer & Wild Turkey Study Group Meeting. September 2014. Potosi, Missouri.
2010	A potential model framework to investigate disease progression. In: 2010 White-nose Syndrome Symposium: May 2010, Pittsburgh, Pennsylvania. U.S. Fish and Wildlife Service.
2007	Capacity building of wildlife managers in dealing with wildlife emergencies. Poster presented in the Annual Meeting of the Society for Conservation Biology, Port Elizabeth, South Africa.

MEMBERSHIPS AND PROFESSIONAL ACTIVITIES

Memberships

2017 - current	Full member & reviewer, CoMSES Net (Network for Computational Modeling in Social and Ecological Sciences).
2016 - current	Certified EpiCore member. EpiCore is a community of health professionals around the world, providing expertise to verify suspected or rumored disease outbreaks.
2024	The Wildlife Society

Associate Editor / Subject Editor

2026 – current	Editorial Board Member, International Journal of Wildlife Health and Conservation.
2025 – current	Associate Editor, Ecological Solutions and Evidence, British Ecological Society.
2022 - current	Review Editor, Models in Ecology and Evolution (specialty section of Frontiers in Ecology and Evolution).
2014 - current	Journal of Threatened Taxa
2016, 2017	Oecologia Australis

Grant reviewer

2026	NIFA-USDA A1181 Agricultural Biosecurity Panel
2026	Scientific Advisory Board, Morris Animal Foundation (First Award and Fellowship Training proposals)
2025	Legislative-Citizen Commission on Minnesota Resources (LCCMR). Reviewer for projects recommended to the state legislature for funding in 2026.
2024	French National Research Agency (ANR) 2024 CE35 (Infectious Disease & Environment)
2024	NIFA-USDA A1181 Agricultural Biosecurity Panel
2024	Morris Animal Foundation, Rapid response review Panel
2023	Scientific Advisory Board, Morris Animal Foundation (First Award and Fellowship Training proposals)
2023	National Science Foundation (Division of Environmental Biology)
2022	National Science Foundation (Human-Environment and Geographical Sciences Program)

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Research, Qeios, Science of the Total Environment, Scientific Reports, The Lancet Regional Health – Southeast Asia, Transboundary and Emerging Diseases, Tropical Animal Health and Production, Vaccine, Veterinary Medicine & Science, Veterinary Parasitology, Urban Ecosystems, Ursus, Zoonoses and Public Health.